

CO1-AT3-Coding Challenge

View Only

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QUESTIONS

Q1

Smart University Payroll
Management System
25 marks

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Smart University Payroll Management System

25
marks

A university employs three categories of staff:
Teaching Faculty, Technical Staff, and Visiting Faculty.
The payroll system should calculate the monthly



SIMATS
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JAVA PROGRAMMING



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teachingHours incentivePerHour Salary: basicSalary +
(teachingHours × incentivePerHour) TechnicalStaff
overtimeHours overtimeRate Salary: basicSalary +
(overtimeHours × overtimeRate) VisitingFaculty
lecturesHandled paymentPerLecture Salary:
lecturesHandled × paymentPerLecture Override
calculateSalary() in all subclasses. Create an array of
Employee objects and demonstrate runtime
polymorphism. Display: Employee details Monthly
salary Highest-paid employee Average salary Search
an employee using the employee ID.

Test Case 1 (Normal Case)

Input

3

TeachingFaculty

101

Dr. Arun

CSE

50000

20

500

TechnicalStaff

102

Mr. Ravi

ECE

35000

15

300

VisitingFaculty

103

Dr. Kumar

IT

0

40

1200

Search Employee ID:

102

Expected Output

Employee Details

Employee ID : 101

Name : Dr. Arun

Department : CSE

Monthly Salary : 60000.0

Employee ID : 102

Name : Mr. Ravi

Department : ECE

Monthly Salary : 39500.0

Employee ID : 103

Name : Dr. Kumar

Department : IT

Monthly Salary : 48000.0

Highest Paid Employee

Employee ID : 101

Salary : 60000.0

Average Salary

49166.67

Employee Found

Employee ID : 102

Name : Mr. Ravi

Salary : 39500.0

Your Code

```
1 import java.util.*;
2
3 class Main
4 {
5     public static String readInput(Scanner
6     {
7         String s = "";
8
9         while(sc.hasNextLine())
10        {
11            s = sc.nextLine().trim();
12
13            if(!s.isEmpty())
14            {
15                return s;
16            }
17        }
18
19        return s;
20    }
21
22    public static void main(String args[])
23    {
24        Scanner sc = new Scanner(System.i
25
26        int n = Integer.parseInt(readInpu
27
28        Employee emp[] = new Employee[n];
29
30        for(int i=0;i<n;i++)
```

```
31         {
32             String type = readInput(sc);
33
34             int id = Integer.parseInt(readInput(sc));
35             String name = readInput(sc);
36             String department = readInput(sc);
37             double basicSalary = Double.parseDouble(readInput(sc));
38
39             if(type.equalsIgnoreCase("Teacher"))
40             {
41                 int teachingHours = Integer.parseInt(readInput(sc));
42                 double incentivePerHour = Double.parseDouble(readInput(sc));
43
44                 emp[i] = new TeachingFaculty(id, name, department, basicSalary, teachingHours, incentivePerHour);
45             }
46             else if(type.equalsIgnoreCase("Technical Staff"))
47             {
48                 int overtimeHours = Integer.parseInt(readInput(sc));
49                 double overtimeRate = Double.parseDouble(readInput(sc));
50
51                 emp[i] = new TechnicalStaff(id, name, department, basicSalary, overtimeHours, overtimeRate);
52             }
53             else if(type.equalsIgnoreCase("Visiting Faculty"))
54             {
55                 int lecturesHandled = Integer.parseInt(readInput(sc));
56                 double paymentPerLecture = Double.parseDouble(readInput(sc));
57
58                 emp[i] = new VisitingFaculty(id, name, department, basicSalary, lecturesHandled, paymentPerLecture);
59             }
60         }
61     }
62
63     String search = readInput(sc);
64
65     if(search.equalsIgnoreCase("Search"))
66     {
67         search = readInput(sc);
68     }
69
70     int searchId = Integer.parseInt(search);
71
72     Employee highest = emp[0];
73     double totalSalary = 0;
74
75     System.out.println("Employee Data");
76     System.out.println();
```

```
79
80     for(int i=0;i<n;i++)
81     {
82         emp[i].displayEmployee();
83
84         System.out.println("Monthly S
85         System.out.println();
86
87         totalSalary += emp[i].calcula
88
89         if(emp[i].calculateSalary() >
90         {
91             highest = emp[i];
92         }
93     }
94
95     System.out.println("Highest Paid
96     System.out.println("Employee ID :
97     System.out.println("Salary : " +
98     System.out.println();
99
100    System.out.println("Average Salar
101
102    double avg = totalSalary / n;
103
104    if(avg == Math.floor(avg))
105        System.out.println(avg);
106    else
107        System.out.printf("%.2f%n", av
108
109    System.out.println();
110
111    boolean found = false;
112
113    for(int i=0;i<n;i++)
114    {
115        if(emp[i].getEmployeeId()==se
116        {
117            System.out.println("Emplc
118            System.out.println("Emplc
119            System.out.println("Name
120            System.out.println("Salar
121
122            found = true;
123            break;
124        }
125    }
126
```

```
127         if(!found)
128         {
129             System.out.println("Employee
130         }
131     }
132 }
133
134 class Employee
135 {
136     private int employeeId;
137     private String employeeName;
138     private String department;
139     private double basicSalary;
140
141     public Employee()
142     {
143
144     }
145
146     public Employee(int employeeId,String
147                     String department,dou
148     {
149         this.employeeId=employeeId;
150         this.employeeName=employeeName;
151         this.department=department;
152         this.basicSalary=basicSalary;
153     }
154
155     public int getEmployeeId()
156     {
157         return employeeId;
158     }
159
160     public String getEmployeeName()
161     {
162         return employeeName;
163     }
164
165     public String getDepartment()
166     {
167         return department;
168     }
169
170     public double getBasicSalary()
171     {
172         return basicSalary;
173     }
174
```

```
175     public double calculateSalary()
176     {
177         return basicSalary;
178     }
179
180     public void displayEmployee()
181     {
182         System.out.println("Employee ID :
183         System.out.println("Name : " + em
184         System.out.println("Department :
185     }
186 }
187 class TeachingFaculty extends Employee
188 {
189     private int teachingHours;
190     private double incentivePerHour;
191
192     public TeachingFaculty()
193     {
194
195     }
196
197     public TeachingFaculty(int employeeId
198                             String departm
199                             int teachingHc
200     {
201         super(employeeId, employeeName, dep
202         this.teachingHours = teachingHour
203         this.incentivePerHour = incentive
204     }
205
206     @Override
207     public double calculateSalary()
208     {
209         return getBasicSalary() + (teachi
210     }
211 }
212
213 class TechnicalStaff extends Employee
214 {
215     private int overtimeHours;
216     private double overtimeRate;
217
218     public TechnicalStaff()
219     {
220
221     }
222 }
```

```
223     public TechnicalStaff(int employeeId,
224                             String departme
225                             int overtimeHou
226     {
227         super(employeeId, employeeName, dep
228         this.overtimeHours = overtimeHour
229         this.overtimeRate = overtimeRate;
230     }
231
232     @Override
233     public double calculateSalary()
234     {
235         return getBasicSalary() + (overti
236     }
237 }
238
239 class VisitingFaculty extends Employee
240 {
241     private int lecturesHandled;
242     private double paymentPerLecture;
243
244     public VisitingFaculty()
245     {
246
247     }
248
249     public VisitingFaculty(int employeeId
250                             String departm
251                             int lecturesHa
252     {
253         super(employeeId, employeeName, dep
254         this.lecturesHandled = lecturesHa
255         this.paymentPerLecture = paymentP
256     }
257
258     @Override
259     public double calculateSalary()
260     {
261         return getBasicSalary() + (lectur
262     }
263 }
```



Input

3

TeachingFaculty
101



Output

Employee Details

Employee ID : 101
Name : Dr.Arun
Department : CSE
Monthly Salary : 60000.0



✔ Submitted

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